ZOOLOGY

CLASSIFICATION OF ANIMAL KINGDOM

A. Phylum Protozoa

Main features

i. Its body is made of only one cell.
ii. There are one or more nuclei in its cytoplasm.
iii. These are of both types–commensalism and parasite.
iv. All the metabolic activities (eating, digestion, respiration, excretion, reproduction) take place in unicellular body.
v. Respiration and excretion take place by diffusion.

Example: Amoeba and Euglena.

B. Phylum Porifera

Main features

All animals of this group are found in water and bear pores in the body.

i. These are multicellular animals but cells do not make regular tissues.
ii. Numerous pores known as ostia found on body wall.
iii. Skeleton is made up of minute calcareos or silicon spicules.

Examples: Sycon, Sponge, etc.

C. Phylum Coelenterata

Main features

i. Animals are aquatic and diploblastic.
ii. Around the mouth some thread-like structures are found known as tentacles.
iii. Body radial symmetry.
iv. Specialised cnidoblast cells are found which help in catching the food.

Example: Hydra, Jelly fish.

D. Phylum Platyhelminthes

Main features:

i. Triploblastic and no body cavity.
ii. Dorsoventrally flattened animal.
iii. Alimentary canal with single opening, anus absent.
iv. Excretion takes place by flame cells.
v. There is no skeleton, respiratory organ, circulatory system, etc.
vi. These are hermaphrodite animals.

Example: Planaria.

E. Phylum Aschelminthes

Main features

i. Long, cylindrical, unsegmented worm.
ii. Bilaterally symmetrical and triploblastic.
iii. Alimentary canal is complete in which mouth and anus both are present.
iv. There is no circulatory system and respiratory system but nervous system is developed.
v. Excretion takes place through Protonephridia.
vi. They are unisexual.

Example: Roundworm.

☞ **Note:** Enterobius (pinworm/threadworm): It is found mainly in the anus of a child. Filarial disease is caused by *Wuchereria bancrofti*.

F. Phylum Annelida

Main features

i. Body is long, thin, soft and metamerically segmented.
ii. Locomotion takes place through Setae made up of Chitin.
iii. Alimentary canal is well developed.
iv. Nervous system is normal and blood is red.
v. Excretion by nephridia.
vi. Both unisexual and bisexual.

Example: Earthworm.

☞ **Note:** There are four pairs of heart in earthworm.

G. Phylum Arthropoda

Main features

i. Body is divided into three parts– Head, Thorax and Abdomen.
ii. Body is covered with a thick chitinous exoskeleton.
iii. Jointed leg.
iv. Circulatory system is of open type.
v. Its body cavities are called haemocoels.
vi. Trachea, book lungs, body surface are respiratory parts.
vii. These are mainly unisexual and fertilization takes place inside the body.

Example: Cockroach and Prawn.

i. There are six feet and four wings in insects.
ii. There are 13 chambers in a cockroach's heart.

H. Phylum Mollusca

Main features

i. Body is divided into head and muscular foot.
ii. Mantle is always present in it.
iii. Alimentary canal is well developed.
iv. Respiration takes place through gills or ctenidia.
v. Blood is colourless.
vi. Excretion takes place through kidneys.

Example: Pila, Octopus.

I. Phylum Echinodermata

Main features

i. All the animals in this group are marine.
ii. Water vascular system is present.
iii. There are tube feet for locomotion, taking food which works as sensational organ.
iv. Brain is developed in nervous system.
v. There is a special capacity of regeneration.

Examples: Star fish, Sea urchin.

J. Phylum Chordata

Main features

i. Notochord is present in it.
ii. All the chordates are triploblastic, coelomate and bilaterally symmetrical.
iii. A dorsal hollow tubular nerve cord and paired pharyngeal gill slits are other features of chordates.

SOME MAIN GROUPS OF PHYLUM CHORDATA

PISCES

Main features

i. Aquatic life.
ii. All these are cold blooded animals.
iii. Heart pumps only impure blood and have two chambers.
iv. Respiration takes place through gills.

Example: Hippopotamus.

AMPHIBIA

i. These are found both on land and water. All these are cold-blooded.
ii. Respiration takes place through gill, skin and lungs.
iii. They have three chambered heart.

Example: Frog, Necturus, Toad, Ichthyophis, Salamander.

REPTILIA

Main features

i. Land vertebrate, cold-blooded, terrestrial or aquatic vertebrates.

ii. Respiration takes place through lungs.
iii. Its eggs are covered with shell made up of calcium carbonate.

Example: Lizard, snake, tortoise.

☞ **Note:** Mesozoic era is called the era of reptiles.

Cobra is the only snake which makes nests.
Heloderma is the only poisonous lizard.
Sea snake which is called Hydrophis is the world's most poisonous snake.

AVES

Main features

i. The animals of this group are warm-blooded tetrapod vertebrates with flight adaptation.
ii. Their four-feet are modified into wings to fly.
iii. They respire through lungs.

Example: Crow, Peacock, Parrot, etc.

MAMMALIA

Main features

i. These are warm-blooded animals.
ii. There is no nucleus in their red blood cells (except in camel and lama).
iii. Skin of mammals has hair.
iv. Mammalia is divided into three sub-classes:
  i. **Prototheria:** It lays eggs, e.g., Echidna.
  ii. **Metatheria:** It bears the immature child, e.g., Kangaroo.
  iii. **Eutheria:** It bears the well developed child, e.g., humans.

Types of Vertebrates

Jawless fishes (Agnatha)	Hagfish, lamprey
Cartilaginous fishes	Sharks, skates, rays, chimaeras
Bony fishes	Sturgeon, herring, salmon, perch, cod, coelacanth
Amphibians	Frogs and toads, salamanders, newts, caecilians
Reptiles	Snakes, crocodiles, alligators, lizards, turtles, tortoises
Birds (Aves)	Penguin, flamingo, eagle, turkey, thrush, parrot
Mammals	Platypus, kangaroo, bat, lion, wolf, mouse, seal, antelope, cow, dolphin, whale, lemur, monkey, ape, human

ANIMAL TISSUE

i. **Epithelial Tissue:** Epithelial tissue cover the external surface of the body and internal free surface of many organs.
Example: skin, intestine, gland.
ii. **Connective Tissue:** These tissues connect and bind different tissues and organs.
Example: Adipose tissue found beneath the skin. Ligament is made up of fibrous connective tissue, cartilage, bone and blood.

☞ **Note:** Blood is only tissue which is found in the form of fluid.

iii. **Muscular Tissue:** This is also known as contractile tissue. All the muscles of the body are made up of this tissue.
(a) **Unstriped:** This muscle tissue is found on the walls of those parts which are not controlled by will.
(b) **Striped:** These muscles are found in the parts of the body that move voluntarily.
(c) **Cardiac:** These muscles are found only on the walls of the heart.

The largest muscle of the human body is Gluteus Maximus.
The smallest muscle of the human body is Stapedius.

iv. **Nervous Tissue:** This tissue is also called sensitive tissue.

HUMAN BLOOD

- The quantity of blood in the human's body is 7% of the total weight.
- Blood is fluid connective tissue and composed of blood corpuscles, plasma and platelets.
- It is slightly alkaline in nature (pH 7.4).
- Its volume in an adult is 5.8 L.

- People who live at high altitudes have more blood than those who live in low regions. This extra blood supplies additional oxygen to body cells.
- During blood clotting fibrinogen changes into fibrin by thrombin which is obtained from thromboplastin in the presence of Ca^{2+}.
- Female contains half litre of blood less in comparison to male.

BLOOD CONSISTS OF TWO PARTS

(A) Plasma; and (B) Blood corpuscles.

(A) **Plasma:** This is the liquid part of blood. 60% of the blood is plasma. Its 90% part is water, 7% protein, 0.9% salt and 0.1% is glucose.
- **Function of plasma:** Transportation of digested food, hormones, excretory product, etc. from the body takes place through plasma.
- **Serum:** When Fibrinogen and protein is extracted out of plasma the remaining plasma is called serum.

(B) **Blood corpuscles:** This is the remaining 40% part of the blood.

i. **Red Blood Corpuscles (RBC):** Red Blood Corpuscles (RBC) in mammal is biconcave.
- There is no nucleus in it. Exception–Camel and Lama. RBC is formed in Bone marrow.
- Its life span is from 20 days to 120 days.
- Its destruction takes place in liver and spleen. Therefore, liver is the grave of RBC.
- It contains haemoglobin, in which haeme iron containing compound found and due to this the colour of blood is red.
- The main function of RBC is to carry oxygen to all cells of the body bring back the carbon dioxide.

ii. **White Blood Corpuscles (WBC) or Leucocytes:**
- Its formation takes place in Bone marrow, lymph node and sometimes in liver and spleen.
- Its life span is from 1 to 2 days.
- Nucleus is present in the White Blood Corpuscles.
- Its main function is to protect the body from the disease. The ratio of RBC and WBC is 600 : 1.

iii. **Blood Platelets or Thrombocytes:** It is found only in the blood of human and other mammals.
- There is no nucleus in it.
- Its formation takes place in Bone marrow.
- Its life span is from 3 to 5 days.
- It dies in the Spleen.
- Its main function is to help in clotting of blood.

FUNCTIONS OF BLOOD

i. To control the temperature of the body and to protect the body from diseases.
ii. Clotting of blood.
iii. Transportation of O_2, CO_2, digested food, conduction of hormones, etc.
iv. To help in establishing coordination among different parts.

- The main reason behind the difference in blood of human is the glycoprotein which is found in Red Blood Corpuscles called antigens. Antigens are of two types: Antigen A and Antigen B.
- On the basis of presence of Antigen or Glycoprotein, there are four group of blood in human:
  (a) That contains Antigen A–Blood Group A.
  (b) That contains Antigen B–Blood Group B.
  (c) That contains both the Antigens A and B–Blood Group AB.
  (d) That contains neither of the Antigens–Blood Group O.

An opposite type of protein is found in blood plasma. This is called antibody. This is also of two types–Antibody 'a' and Antibody 'b'.

Blood Transfusion: Antigen 'A' and antibody 'a', Antigen 'B' and antibody 'b' cannot live together. In case of so happened these get most sticky, such spoils the blood. This is called agglutination of blood.

Blood Group **O** is called **Universal Donor** because it does not contain any antigen.

Blood Group **AB** is called **Universal Receptor** because it does not contain any antibody.

If in the blood of people it is found, their blood is said to be Rh-positive and if in the blood of people it is not found, their blood is said to be Rh-negative.

At the time of blood transfusion, Rh-factor is also tested. Rh-positive is given to Rh-positive blood and Rh-negative is given to Rh-negative blood only.

Erythroblastosis Fetalis: If the father's blood is Rh-positive and the mother's blood is Rh-negative, then the child to be born dies at pregnancy or short span of time after the birth. (This happens in the case of second issue).

Possible Combinations of Blood Groups

Male	Female	Blood group of Children not possible
A	A	B & AB
A	B	–
A	AB	O
A	O	B or AB
B	B	A, AB
B	AB	O
B	O	A, AB
AB	AB	O
AB	O	O, AB
O	O	A, B, AB

BLOOD PRESSURE (BP)

- The pressure created by the blood on the walls of the blood vessels due to the repeated pumping of heart is called **blood pressure**. It is measured by **sphygmomanometer.**
- Blood pressure in a normal person is 120/80 mm Hg.
- If a person has persistent high blood pressure then it is called hypertension and persistent high blood pressure is 150/90 mm Hg.
- **Hypotension** is condition of low blood pressure, *i.e.,* persistent 100/50 mm Hg.
- Electrocardiograph (ECG) is used to check proper working of heart.

SYSTEM OF THE HUMAN BODY

EPIDERMIS

- The top layer of skin made up of epithelial cells and does not contain blood vessels.

DERMIS

- It gives elasticity to the integument, allowing stretching and conferring flexibility, while resisting distortions, wrinkling and sagging.

HYPODERMIS

- It is made up of adipose tissue.

TEETH

- With the help of teeth the food is chewed. Teeth are of four types
  i. Incisors
  ii. Canines
  iii. Premolars
  iv. Molars
- Hardest part in the body is tooth enamel.

TONGUE

- Saliva, secreted by the salivary glands, is mixed with the chewed food by the tongue.
- Complete digestion process takes place in following four steps:
  i. Ingestion of Food
  ii. Digestion in Mouth.
  iii. Digestion in Stomach.
  iv. Digestion in Intestine.
- The food passes down through the oesophagus into stomach.
- Now food is mixed with gastric juice and hydrochloric acid which disinfect the food and creates acidic medium.
- Pepsin digests proteins and converts them into peptones.
- Renin coverts milk into curd.
- The digested food now is called chyme.
  1. **Ingestion:** Taking the food into the mouth is called ingestion.
  2. **Digestion:** Conversion of non-absorbable food into absorbable form. The digestion of the food is started in the mouth.
- Saliva is secreted by salivary gland in mouth in which two types of enzymes are found, ptyalin and maltase. They convert starch into simple sugar and make it digestible.

- From the mouth the food goes into stomach through foodpipe.
- No digestion takes place in foodpipe.

DIGESTION IN STOMACH

- The food lies approximately for four hours in the stomach.
- Hydrochloric acid secreted from the Oxyntic cells of the stomach kill all the bacteria coming with food and accelerate the reaction of enzymes.
- The enzymes in the gastric juice of stomach are–Pepsin and Renin.
- Pepsin breaks down the protein into peptones.
- Renin breaks down the caseinogen into casein.

DIGESTION IN DUODENUM

As soon as the food reaches the duodenum bile juice from liver combines with it. Bile juice is alkaline and it turns the acidic medium of food into alkaline.

Here, pancreatic juice from pancreas combines with food. It contains three types of enzymes:

(a) **Trypsin:** It converts the protein and peptone into polypeptides and amino acid.

(b) **Amylase:** It converts the starch into soluble sugar.

(c) **Lipase:** It converts the emulsified fats into glycerol and fatty acids.

SMALL INTESTINE

Digestion in Intestine

Food passes into ileum and mixes with intestinal juice, where:

- Maltase converts into glucose.
- Lactose converts into glucose and galactose.
- Sucrose converts into glucose and fructose.
- Trypsin digests the peptides into amino acids.
- Food now is called chyle.

ABSORPTION AND ASSIMILATION OF DIGESTED FOOD

- Ileum's internal surface has finger-like folds called **villi.**
- It helps in absorption of food.

EJECTION OF UNWANTED FOOD

- Digested food passes into large intestine.
- Large intestine cannot absorb food, but absorbs much of the water.
- The remaining semi-solid waste is called **faeces** and is passed into rectum.

The main organs participating in digestion:

Liver: This is the largest gland of the human body. Its weight is approximately 1.5–2 kilogram.

- Bile is secreted through liver only.
- Liver converts excess of amino acid into ammonia by deamination. The ammonia is further converted into urea by ornithine cycle. Urea comes out from body through kidney.
- Liver converts some quantity of protein into glucose during defecation of carbohydrate.
- Liver regulates the quantity of glucose in the blood.
- In case of decrease of fat in food liver converts some of the parts of carbohydrates into fat.
- The production of fibrinogen protein takes place by liver which helps in clotting of blood.
- The production of Heparin protein takes place in liver which prevent the clotting of blood inside the body.
- The liver reserves some quantity of iron, copper and vitamin.
- It helps in regulating the body temperature.
- Liver is an important clue in investigating a person's death that is been due to poison in food.

Gall Bladder: Gall bladder is a pear shaped sac, in which the bile coming out of liver is stored.

- Bile is a yellowish-green coloured alkaline liquid, whose pH value is 7.7.
- The quantity of water is 85% and the quantity of bile pigment is 12% in water.

The main functions of bile are as under:

i. It makes the medium of food alkaline so that pancreatic juice can work.

ii. It kills the harmful bacteria coming with food.

iii. It emulsifies the fats.

iv. It accelerates the bowel movement of intestine by which digestive juices in the food mix well.
v. It is helpful in the absorption of vitamin K and other vitamins mixed in fats.

In case of obstruction in bile duct, liver cells stop taking bilirubin form. As a result, bilirubin spreads throughout the body. This is called jaundice.

Pancreas: This is the second largest gland of the human body. It acts simultaneously as endocrine and exocrine type of gland.

Islets of Langerhans: This is a part of the pancreas.

Insulin: It is secreted by β-Cells of Islets of Langerhans which is a part of pancreas. It controls the process of making glycogen from glucose.

Diabetes is caused due to the deficiency of insulin.

- Excessive flow of insulin causes Hypoglycemia in which one loses the producing capacity and vision deterioration.

Glucagon: It re-converts the glycogen into glucose.

CIRCULATORY SYSTEM

The discovery of blood circulation was done by William Harvey.

HEART

It remains safe in the pericardial membrane. Heart of the human is made up of four chambers.

- The chamber which receives the blood from body tissues is called **auricles** and the chambers of heart which pump blood to body tissues are called **ventricles.**
- There is a thin two layered sac around the heart known as **pericardium,** filled with a watery fluid called pericardial fluid, which allows frictionless movements of heart and protects it from mechanical shocks.
- The blood vessel: carrying the blood from the body towards the heart is called vein.
- In the vein there is impure blood i.e. carbon dioxide mixed blood with the exception is pulmonary vein, which always carry pure blood.
- Pulmonary vein carrying the blood from lungs to left auricle. It has pure blood.
- The blood vessel carrying the blood from the heart towards the body is called artery.
- In artery there is pure blood i.e. oxygen mixed blood. Its exception is pulmonary artery.
- Pulmonary artery carries the blood from right ventricle to lungs. It contains impure blood.
- The artery carrying blood to the muscles of the heart are called coronary arteries. Any type of hindrance in it cause heart attack.
- **A fish** has only two-chambered heart.
- **In amphibians,** heart is three-chambered.
- In **crocodile, birds** and **mammals** the heart is divided into four chambers.

  Course of circulation: Mammals have double circulation. It is because blood have to cross two times from heart before circulating throughout body.
- To pump out blood, the heart chamber undergoes alternate contraction called **systole** and relaxation called **diastole.**
- Arteries carry **pure blood** from the heart while veins carry **impure blood** to the heart.
- Human heartbeat is myogenic in nature, i.e., initiated by a patch of modified heart muscles itself without requiring an external stimulation. This patch is called **SA node** (sino-auricular node) or **pacemaker.**
- The normal **rate of heartbeat** of a newborn baby is about 140 per minute.
- When SA-node becomes defective, i.e., it does not generate cardiac impulse, it can be cured by surgical grafting of an artificial pacemaker (an electric device) in the chest of the patient. It stimulates the heart electrically at regular intervals.
- The blood pressure of a normal human is 120/80.
- Blood pressure is measured by sphygmo manometer.
- Thyroxine and adrenaline are the hormones which independently controls the heartbeat.
- The CO_2 present in the blood accelerates the heartbeat by reducing the pH.

CAPILLARIES

- These are the thinnest blood vessels and connect arteries to the veins.

LYMPH CIRCULATORY SYSTEM

- The light yellow fluid found in the inter-cellular spaces between different tissues and cells is called lymph.
- The corpuscles found in lymph are called lymphocytes.
- Lymph flows only in one direction from tissue towards heart.

Functions of Lymph

i. The lymphocytes present in lymph help prevent the body from disease by killing the harmful bacteria.
ii. Lymph forms the lymphocytes.

EXCRETORY SYSTEM

KIDNEY

- It is bean-shaped, chocolate brown structure lying in the abdomen, one on each side of the vertebral column just below the diaphragm.
- The left kidney is placed in little higher than the right kidney (but reverse in rabbit).
- These form the urine and controls osmotic pressure within the organism with respect to external environment.
- Nephrons are the functional and structural unit of kidney. They contain Bowman's capsule and Henle's loop.
- The process of filtration of liquids into the cavity of Bowman's capsule is called ultrafiltration.
- The main function of the kidneys is purification of blood plasma, i.e. to excrete the unwanted nitrogenous waste substances through urination.
- In the kidneys average 125 ml per minute blood is filtered.

URETERS

- These bring the urine downwards and open into urinary bladder.

URINARY BLADDER

- It temporarily stores the urine.

URETHRA

- In females, this tube is small and serves as a passage of urine only.
- In males, it is long and functions as a common passage for urine and spermatic fluids.

URINE

- It is pale yellow coloured fluid due to presence of urochrome pigment.
- It is acidic in nature (pH 6.0) and is slightly heavier than water.
- **Chemical composition of urine**: Water is 95-96%, urea is 2% and some other substances like uric acid, creatinine, etc. are 2-3%.
- Kidney stone is calcium oxalate.

i. **Skin:** Oil glands and sweat glands found in the skin secrete sebum and sweat.
ii. **Liver:** Liver cells play the main role in excretion by converting and more amino acids and ammonia of blood into urea.
iii. **Lungs:** The lungs excrete two types of gaseous substances–carbon dioxide and water vapour.

Hemodialysis: Process of removal of excess urea from the blood of patient using artificial kidney.

EXCRETORY PRODUCTS

AMMONOTELIC ANIMALS

- These animals excrete nitrogen in the form of ammonia, e.g., aquatic invertebrates.

UREOTELIC ANIMALS

- They excrete nitrogen in the form of urea, e.g., mammal (man).

URICOTELIC ANIMALS

- They excrete the nitrogenous wastes in the form of uric acid, e.g., Reptiles, snakes.

CENTRAL NERVOUS SYSTEM

- Nervous system is found only in animals and absent in plants.
- Part of the nervous system which keeps control on the whole body and on nervous system itself is called Central Nervous System. The Central Nervous System of human is made up of two parts–Brain and Spinal Cord.
- Brain is covered by a membrane called meninges. It is situated in a bony box

called cranium which protects it from external injury.

BRAIN

- Brain lies in the cranium of skull.
- The functions of brain parts are as follows:
  i. **Cerebrum** leads to consciousness, storage of memory of information.
  ii. **Thalamus** deals with pain, pressure and temperature.
  iii. **Hypothalamus** deals with water balance in body, behavioural patterns of sex, sleep, stress emotions, etc. It also regulates pituitary hormones and metabolism of fat, carbohydrate and water.
  iv. **Midbrain** deals with visual analysis, etc.
  v. **Cerebellum** controls coordination of accurate movements and balancing.
  vi. Medulla oblongata is long connecting part of brain to spinal cord. It deals with control of heartbeats, blood vessels, breathing, salivary secretion and most of reflex and involuntary (uncontrolled) movements.

SPINAL CORD

The posterior region of the medulla oblongata is the spinal cord. Its main functions are:

(a) Coordination and control of reflex actions i.e. it works as the centre of the reflex actions.
(b) It carries the waves coming out of the brain.

PERIPHERAL NERVOUS SYSTEM

- Peripheral Nervous System is made up of the nerves arising from brain and spinal cord.
- The unit of nervous tissues is called neuron of nerve cell.

AUTONOMOUS NERVOUS SYSTEM (ANS)

- It is entirely motor and operates without conscious control. Autonomic Nervous System consists of two divisions.
  i. **Sympathetic nervous system** increases defence system of body against adverse conditions.
  ii. **Parasympathetic nervous system** provides relaxation, comfort, pleasure at the time of rest.
- Electroencephalogram (EEG) is a test that measures and records the electrical activity of our brain.

SKELETAL SYSTEM

(i) Axial Skeleton (80 Bones)

- It includes skull, vertebral column and bones of chest.
- Vertebral column is responsible for the upright position of the human body.

(ii) Appendicular Skeleton (126 Bones)

- Their functions are to make locomotion possible and to protect the major organs of locomotion, digestion, excretion, and reproduction.

i. **Skull:** There are 29 bones in it.
ii. **Vertebral Column:** The vertebral column of the human is made up of 33 vertebrae.

FUNCTIONS OF VERTEBRAL COLUMN

i. Holds the head.
ii. It provides the base to the neck, and body.
iii. It helps the human in standing, walking, etc.
iv. It provides flexibility to the neck and body by which a human can move his neck and body in any direction.
v. It provides protection to spinal cord.

FUNCTIONS OF THE SKELETAL SYSTEM

i. To provide a definite shape to the body.
ii. To provide protection to soft parts of the body.
iii. To provide a base to the muscles for joining.
iv. To help in respiration and nutrition.
v. To form Red Blood Corpuscles.

- The total number of bones in a human's body–206
- The total number of bones during childhood–300

- The largest bone of the body–Femur (bone of thigh).
- The smallest bone of the body–Stapes (bone of ear).

☞ **Note:** The muscles and bones are joined together by tendon. The muscle which joins bone to bone is called ligament.

DISEASES OF SKELETAL SYSTEM

- Hard tissue deposits over articular cartilage along with higher secretion of synovial fluid causing pain and stiffness lead to **rheumatoid arthritis**.
- Tearing of articular cartilage and development of bony lumps at places causing pain, stiffness and permanent bending lead to **osteoarthritis.**
- **Osteoporosis** is loss of bone density due to excessive absorption of calcium and phosphorus from the bone.
- Osteoporosis is a hereditary disease marked by abnormally dense bone, and by the common occurrence of fractures of affected bone.
- Tendons join the muscles and bones.
- The muscles which join bone-to-bone are called ligaments.

MUSCULAR SYSTEM

- Human body has about 639 types of muscles.

ENDOCRINE SYSTEM

(a) **Exocrine glands:** Glands which have duct are called exocrine glands. Secretion of enzymes pass through it. **Example:** Lactic gland, Sweat gland.

(b) **Endocrine glands:** These are ductless glands. Hormones are secreted by these glands. Hormones are sent to the different parts of the body through blood plasma. **Example:** Pituitary gland, Thyroid gland.

PITUITARY GLAND

i. **STH hormone (Somatotropic Hormone):** It controls the growth of the body especially the growth of bones. By the excessiveness of STH gigantism and acromegaly are caused, in which height of the human grows abnormally. Lack of STH causes dwarfism in human.

ii. **TSH hormone (Thyroid Stimulating Hormone):** It stimulates its thyroid gland to secrete hormone.

iii. **ACTH hormone (Adrenocorticotropic Hormone):** It controls its secretion of adrenal cortex.

iv. **GTH hormone (Gonadotropin Hormone):** It controls the function of gonads.

(a) **FSH Hormone (Follicle-stimulating Hormone):** In males, it stimulates Sertoli Cells to genesis in the seminiferous tubules of the testis. In female, it stimulates the Graafian follicles of the ovary to secret the hormone oestrogen.

(b) **LH Hormone (Luteinizing Hormone):** Also called Interstitial cell-stimulating hormone (ICSH). Secretion of testosterone hormone takes place in male and in case of female estrogen hormone secreted.

v. **LTH Hormone (Lactogenic Hormone):** Its main function is to stimulate secretion of milk in breasts for infants.

vi. **ADH Hormone (Antidiuretic Hormone):** It causes increase in blood pressure. It is helpful in maintaining the water balance in the body and reducing the volume of urine.

Classification of Hormones

Amines	Peptide hormones	Steroids/sterols	Lipids
Adrenaline	Acth Or Corticotropin	Cortisol	Prostaglandins
Dopamine	Vasopressin	Aldosterone	Leukotrienes
Noradrenaline	Calcitonin	Testosterone	Prostacyclin

Melatonin	Corticotropin-Releasing Hormone (Crh)	Androstenedione	Thromboxane
Serotonin	Erythropoietin (Epo)	Oestrogen	
Thyroxine	Follicle-Stimulating Hormone (Fsh)	Estradiol	
Triiodothyronine	Gastrin	Progesterone	
	Glucagon	Progestins	
	Gonadotropin-Releasing Hormone (Gnrh)	Calcitriol	
	Growth Hormone-Releasing Hormone (GHRH)	(Sterol)	
	Insulin		
	Leptin		
	Luteinizing Hormone (LH)		
	Oxytocin		
	Parathyroid Hormone (PTH)		
	Prolactin (PRL)		

THYROID GLAND

- The hormones secreted by it are Thyroxine and Triiodothyronine. Iodine is secreted in more quantity.

FUNCTIONS OF THYROXIN

i. It increases the speed of cellular respiration.
ii. It is necessary for the normal growth of the body particularly for the development of bones, hair, etc.
iii. The normal function of reproductive organs depends on the activeness of thyroid gland.
iv. It controls the water balance of the body in coordination with the hormones of pituitary gland.

DISEASES CAUSED BY THE DEFICIENCY OF THYROXIN

i. Cretinism
ii. Myxedema
iii. Hypothyroidism
iv. Goitre

Diseases caused by the Excessiveness of Thyroxin: Exophthalmic Goitre.

PARATHYROID GLAND

i. **Parathyroid hormone:** This hormone is secreted when there is a deficiency of calcium in the blood.
ii. **Calcitonin:** This hormone is released when there is excess of calcium in the blood.

Hence, hormone secreted by parathyroid gland controls the quantity of calcium in blood.

PANCREAS

- It is both an exocrine and endocrine gland. The islet of Langerhans (endocrine) has three major types of cells:
  i. **alpha-cells** secrete glucagon hormone.
  ii. **beta-cells** secrete insulin,
  iii. **delta-cells** secrete somatostatin.

ADRENAL GLAND

- It is also known as **emergency glands.**
  (a) Cortex (b) Medulla

OVARIAN HORMONES (OESTROGENS)

- It helps in the development of primary and secondary sexual characters (oestradiol, oestriol and estrone).

TESTICULAR HORMONES (ANDROGENS)

- It stimulates growth, maturation and maintenance of male gonads and development of secondary sexual characters, e.g., testosterone and sterone, etc.

PINEAL GLAND

- It is situated in the brain and also known as **clockwork gland.**
- It regulates the ovaries and has an effect on the biological rhythm.
- BCG (Bacillus Calmette Guerin) vaccine is given to protect against TB (Tuberculosis).
- DPT (diphtheria, Pertussis and Tetanus) vaccine is given to babies within first 6 weeks of their birth.
- Fishes like cat fish, gambusia and aquatic birds eat mosquito larvae.

VACCINATION

- It is the process of artificial introduction of germs or the germ substance called **antigen** into the body for developing resistance to a particular disease.
- A vaccine is dead or weakened microbes.

HORMONES SECRETED BY MEDULLA

i. Epinephrine
ii. Nor epinephrine

- The work of both the hormones is similar. These equally increase the relaxation and contraction of heart muscles. As a result, blood pressure increases.
- In case of sudden stop of heartbeat, epinephrine is helpful in re-starting the heartbeat.
- The hormone secreted by Adrenal gland is called fight flight, fright fight hormone.

REPRODUCTIVE SYSTEM

- The process by which new individuals are produced from their parents is called **reproduction.**
- In asexual reproduction, only one parent is involved and sex cells are not involved.
- In **sexual reproduction,** two parents are involved and formation and fusion of gametes takes place.
- Males can produce spermatozoa (sperm) throughout their life from age of 13-14 years.
- If sperm is present, the egg will be fertilized in the fallopian tube.
- After maturity the ovary releases an ovum (egg cell) after every 28 days.
- The connection between developing embryo and mother is by **placenta.** It supplies blood, etc.
- The embryo develops for nine months in uterus. It is called **gestation period**.

GONADS

Ovary: The following hormones are secreted by this:

i. **Estrogen:** It completes the increase of oviduct.
ii. **Progesterone:** It stimulates the thickening of uterus lining during ovarian cycle.
iii. **Relaxin:** During pregnancy it is found in uterus and placenta. This hormone smoothens the pubic symphysis and it widens the uterine cervix, so that the child is delivered easily.

Testes: The hormone secreted by it is called testosterone. It motivates the sexual behaviour and growth of secondary sexual characters.

MENSTRUAL CYCLE

- Reproductive period of a human female extends from puberty (10-14 years) to menopause (40-50 years).
- Menopause is stopping of ovulation and menses.
- The periodic vaginal bleeding may be suppressed during pregnancy, during lactation and permanently stops at menopause.
- Menstrual cycle is controlled by FSH, LH, oestrogen and progesterone.

BIRTH CONTROL METHODS OR CONTRACEPTION

- The prevention of union of sperm and ovum is known as **contraception**. The various methods used for it are diaphragm, contraceptive pills, tubectomy, vasectomy, copper-T, etc.

  Amniocentesis or amniotic fluid test is technique of finding out sex and disorder of foetus.

RESPIRATORY SYSTEM

- Respiration is a catabolic process in which the respired oxygen is used in the oxidation of food resulting in the release of energy.

HUMAN RESPIRATORY SYSTEM

- Overall passage of air in humans is as follows: Nostrils–Pharynx–Larynx–Trachea–Bronchi–Bronchioles – Alveoli–Cells–Blood.

EXTERNAL RESPIRATION

- It involves inspiration and expiration of air.
- **Inspiration** is the process of intake of air. During inspiration, muscles of the diaphragm contract and diaphragm flatten. The lower ribs are raised upward and outwards, the chest cavity enlarges, the air pressure in the lungs is decreased, air rushes into the lungs.
- **Expiration** is breathing out of air. During expiration, relaxation of muscles of the ribs and diaphragm takes place. Diaphragm again becomes dome-shaped. Chest cavity is reduced and air is forced outward through nose and trachea.

INTERNAL RESPIRATION (OXIDATION OF FOOD)

- It is a complex process in which food is broken down to release energy. It is a biochemical phase takes place inside the cell.
- Transportation of oxygen takes place by haemoglobin of blood, whereas transportation of only 10-20% carbon dioxide takes place by haemoglobin of blood.
- Respiration being a catalytic process also **reduces the weight** of the body.

  Glucose is oxidised by oxygen reached into the cell. This process is called cellular respiration.

TYPES OF RESPIRATION

Aerobic Respiration

- The respiration which takes place in the presence of oxygen is known as aerobic respiration.
- In this process, in fact, each glucose molecule is converted into two molecules of pyruvic acid by the process, called **glycolysis.** It takes place in the cytoplasm of the cell. The pyruvic acid formed, releases energy with the formation of carbon dioxide and water (in Krebs' cycle which occur in mitochondria).

Anaerobic Respiration

- The respiration which takes place in the absence of oxygen is known as anaerobic respiration.
- In this process, the respiratory substances are incompletely oxidized to carbon dioxide and alcohol.
  i. **Glycolysis:** Its study was first done by Embden-Meyerhof Pathway. Therefore, it is also called EMP path.
     - Glycolysis is present in both types of respiration, Aerobic and Anaerobic. This process takes place in cytoplasm.
     - As a result of decomposition of one glucose atom in glycolysis two atoms of pyruvic acid are formed.
     - There is no need of oxygen in glycolysis.
  ii. **Krebs' Cycle:** This is also called Citric Acid Cycle or Tricarboxylic Cycle.
     - This process is completed inside the mitochondria in the presence of specific enzymes.
     - Two atoms of each ADP and ATP are formed.
     - In our system maximum ATP atoms are formed during Krebs' Cycle.

  Carbohydrate, fat and protein are the main respiratory substances. At first, oxidation of glucose takes place, then fat. After the consumption of carbohydrate and fat, oxidation of protein start.

SENSE ORGANS

EYE

It consists of three parts.

1. Sclerotic Layer

- Cornea
- Conjunctiva

2. Choroid Layer

It is the middle layer and consists of:

i. **Pupil:** It changes size as the amount of light changes.

ii. Ciliary body.
iii. **Iris:** It controls the amount of light that enters the eye by changing the size of the pupil.
iv. Lens is a biconvex transparent circular solid part located just behind the iris.

3. Retina

- Light sensitive tissue that lines the back of the eye.
- The image formed on retina is real and inverted.
- Rods are highly sensitive to dim light and contain a reddish purple pigment called rhodopsin.
- Cones are sensitive to bright light, hence differentiate the colours.
- The **fovea centralis** is the area of sharpest vision.
- The **blind spot:** no image is formed in this region.

Eye Defects

- **Nearsightedness** (Myopia)
- **Farsightedness** (Hypermetropia)

Astigmatism
Presbyopia

- **Conjunctivitis**

EAR

- Human ear can list in the sound of 60-80 decibel.
- Defects of ear are: **Otalgia ear-ache** (Pain in ear); **Otitis media** (acute infection of middle ear), **labyrinthine** disease (malfunction of inner ear).

NOSE (OLFACTORY ORGAN)

Olfactory cells

- Dos have an acute olfactory sense.

NUTRIENTS

- These are metals, non-metals and their salts other than the four elements–carbon, hydrogen, nitrogen and oxygen and constitute about 4% of total body weight.
- Milk, eggs, meat, fruit, food, vegetables, etc. are the sources of minerals.

Nutrition is one of the basic functions of life in which intake of food, digestion, absorption, assimilation are included.

CARBOHYDRATES

Carbohydrates are organic compounds in which the ratio of Carbon, Hydrogen and Oxygen is 1: 2: 1.

Carbohydrates are classified into three major groups:

(a) **Monosaccharides:** These are the simple sugar made up of single polyhydroxy or ketone unit. Most abundant monosaccharides found in nature are glucose. Triose, tetrose, pentoses, heptoses are the type of monosaccharides.

(b) **Oligosaccharides:** When 2 to 10 monosaccharides join together they form oligosaccharides. Maltose, sucrose, lactose are disaccharides made up of two monosaccharides.

(c) **Polysaccharides:** These are the compounds of sugar which are formed due to joining large number of monosaccharide. Some examples of polysaccharides are starch, glycogen, cellulose, chitin, etc.

Functions of Carbohydrates

1. Carbohydrate works as fuel during the process of respiration, glucose break into CO_2 and H_2O with the release of energy. One gram of glucose gives 4.2 kilo calories energy.
2. Nucleic acids are polymers of nucleosides and nucleotides and contain pentose sugar.
3. Lactose of milk is formed from glucose and galactose.
4. Glucose is used for the formation of fat and amino acid.
5. Carbon skeleton of monosaccharides is used in the formation of fatty acid, chitin, cellulose, etc.

Sources of Carbohydrates

Wheat, rice, maize, sweet potato and other plant and animals are the sources of carbohydrate.

PROTEINS

- This is a complex organic compound made up of 20 types of amino acids.
- Nitrogen is present in protein in addition to C, H and O.
- Twenty-two types of protein is necessary for human body, out of which 12 are synthesized by body itself and remaining 10 are obtained by food are called essential amino acid.
- These are the compounds of carbon (C), hydrogen (H), oxygen (O), nitrogen (N) and sulphur (S). These form 15% part of human body.
- Their main sources are groundnuts, soyabean, pulses, fish, etc.

Functions of Proteins

1. It takes part in the formation of cells, protoplasm and tissues.
2. These are important for physical growth. Physical growth hampers by their deficiency. Lack of proteins causes Kwashiorkor and Marasmus diseases in children.
3. In case of necessity they provide energy to the body.
4. They control the development of genetic characters.
5. These are helpful in conduction also.

Kwashiorkor: In this disease hands and legs of children get slimmed and the stomach comes out.

Marasmus: In this disease muscles of children are loosened.

FATS

- Fat is an ester of glycerol and fatty acid.
- Normally fat remains as solid at 20°C temperature, but if it is in liquid form at this temperature, this is called oil.
- 9.3 kilocalorie energy is liberated from 1 gram fat.
- These are also the compounds of carbon (C), hydrogen (H) and oxygen (O).
- Fatty acids are of two types. Saturated and Unsaturated. Saturated fatty acids are found in coconut oil and palm oil, while unsaturated fatty acids are found in fish oil and vegetable oil.
- Excess of saturated fats raises the level of blood cholesterol and may cause arteriosclerosis (hardening of arteries). This may lead to heart attack.

Main Functions of Fat

1. It provides energy to the body.
2. It remains under the skin and prevents the loss of heat from the body.
3. It makes the food material tasty.
4. It protects different parts of the body from injury.

- Due to the lack of fat skin gets dried, weight of the body decreases and the development of the body checked.
- Due to the excessiveness of fat the body gets fatty, heart disease takes place and blood pressure increases.

ROUGHAGE

- Roughage is another term for dietary fibre, e.g., natural food, *dalia*, etc.
- Helps in retaining water in the body.

VITAMINS

It was first invented by FG Hopkins. However, the term vitamin was coined by **C Funk.**

- They provide no calories, they only regulate chemical reactions occurring in the metabolism of the body.
  - i. **Vitamin soluble in water:** Vitamin-B and Vitamin-C.
  - ii. **Vitamin soluble in fat:** Vitamin-A, Vitamin-D, Vitamin-E and Vitamin-K.
- Cobalt is found in Vitamin-B12.
- Synthesis of vitamins cannot be done by the cells and it is fulfilled by the vitamin foods.
- However, synthesis of Vitamin-D and K takes place in our body.
- Synthesis of Vitamin-D takes place by the ultraviolet rays present in the sunlight through cholesterol (ergesterol) of skin.
- Vitamin-K is synthesized in our colon by the bacteria and from there it is absorbed.

Vitamin	Chemical Name	Solubility	Deficiency Disease	Food Sources
Vitamin A	Retinol	Fat	Night blindness	Orange, ripe, yellow fruits, leafy vegetables, carrots, pumpkin, fish, soymilk, milk
Vitamin B_1	Thiamine	Water	Beriberi	Pork, oatmeal, brown rice, vegetables, potatoes, liver, eggs
Vitamin B_2	Riboflavin	Water	Ariboflavinosis, Glossitis	Dairy product, bananas, popcorn, green beans
Vitamin B_3	Niacin, Nicotinomide	Water	Pellagra	Meat, fish, eggs, mushrooms, seeds, nuts
Vitamin B_5	Pantothenic	Water	Parestheria	Meat, broccoli, avocados
Vitamin B_6	Pyridoxine	Water	Anemia	Meat, true nuts, bananas
Vitamin B_7	Priotin	Water	Dematitis, enteritis	Raw egg yolk, liver, peanuts, leafy green vegetables
Vitamin B_g	Folic acid	Water	Megalobastic anemia	Leafy vegetables, pasta, bread, coreal.
Vitamin B_{12}	Cyanocobalamin	Water	Pernicious anemia	Meat, poultry, fish, eggs, milk
Vitamin C	Ascorbic acid	Water	Scurvy	Many fruits and vegetables
Vitamin D	Cholecalciferol	Fat	Rickets	Fish, eggs, liver, mushrooms
Vitamin E	Tocopherols	Fat	Sterility in males and miscarriage in females	Many fruits and vegetables, nuts and seeds
Vitamin K	Phylloquinone	Fat	Bleeding disthesis	Leafy green vegetables, egg yolks

MINERALS
These control the metabolism of body.

WATER
65-75% weight of the body is water.

Main Functions of Water

1. Water controls the temperature of our body by sweating and vaporizing.
2. It is the important way of excretion of the waste substances from the body.
3. Most of the organic chemical reactions in our body are performed through hydrolysis.

BALANCE DIET
That nutrition, in which all the important nutrients for organism are available in sufficient quantity, is called Balance Diet.

- Louis Pasteur discovered the vaccine of Rabies and pasteurization of milk.

☞ **Note:** AIDS–Acquired Immunodeficiency Syndrome.

Elisa Test: Test of HIV Virus (AIDS).

FOOD ADULTERATION
- Indian Standards Institute (ISI) Mark and Agmark (Agricultural Marketing) are given by the Bureau of Indian Standards after testifying the purity and quality of food.

DISEASES

DISEASES CAUSED BY PROTOZOA:
i. **Diarrhoea:** The reason of this disease is the presence of internal protozoa,

namely *Entamoeba histolytica* which is spread through houseflies.

ii. **Filaria:** This disease is caused by *Wuchereia baoncrofti*. This is circulated by the stings of *culex* mosquitoes. This disease is also known as Elephantiasis.

DISEASES CAUSED BY FUNGUS

i. **Asthma:** This spore of the fungi, namely *Aspergillus fumigate* reaches the lungs of the human and constitutes a net-like formation, thus obstructs the function of lungs. This is an infectious disease.

ii. **Athlete's foot:** This disease is caused by the fungi namely *Tenia pedes*.

iii. **Scabies:** This disease is caused by the fungi, namely *Acarus scably*.

iv. **Baldness:** This is caused by the fungi, namely *Taenia capitis*.

v. **Ringworm:** This disease spreads through the fungi namely *Trycophyton lerucosum*. This is an infectious disease.

SOME OTHER DISEASES

- **Paralysis or Hemiplegia:** In this disease, within a few minutes, a part of the body is paralysed.
  The reason of this disease is due to high blood pressure bursting of any nerve of brain or insufficient supply of blood to brain.
- **Allergy:** Itching, pimples, swelling in body, black spot, eczema, etc. are the examples of allergy.
- **Schizophrenia:** This is a mental disease which usually found in youth. Lectropathy is helpful in this disease.
- **Epilepsy:** This disease is caused by the internal disturbance of brain.
- **Diplopia:** This disease is caused by the paralysis of muscles of the eyes, in which double image is formed.
- **Bronchitis:** It is caused by the inflammation of tubes leading from the windpipe to lungs.
- **Cold:** This is a highly infectious disease and is caused by a virus, which results in bad throat, headache and watery nose.
- **Colic:** Severe pain in the abdomen caused by spasm of the internal organs usually the intestines.
- **Delirium:** It is a serious mental disturbance.
- **Hydrophobia:** A disease caused by bite of a mad dog.
- **Leukaemia:** There is a great increase in the number of white blood corpuscles in system. Swelling of spleen takes place. Death occurs within a few days.
- **Migraine:** An allergic disease in which there is a periodic attack of headache. It is an incurable disease.
- **Obesity:** Excessive fatness is called obesity.
- **Piles:** There are various veins in the rectum. Due to extra pressure in the vein, it prevents the free flow of blood, thus creating problems.
- **Rheumatism:** The symptom of this disease is fever with joints pain.
- **Atherosclerosis:** Deposition of cholesterol particles in the lumen of arteries which prevent the flow of blood is called atherosclerosis.
- **Arteriosclerosis:** Due to deposition of cholesterol and calcium salt, arteries become stiff and rigid. It loses the property of elasticity due to which the wall of arteries may get affected.
- **Uremia:** Presence of excess of urea in blood is called uremia. This is caused by malfunctioning of kidney.
- **Glycosuria:** Presence of excess of glucose in urine is known as glycosuria.
- **Arthritis:** It is a disease in which inflammation of joints takes place.
- **Osteoporosis:** It is an age dependent disorder of bone in which low bones mass and increased fragility takes place.
- **Pneumonia:** Acute inflammation of alveoli of lungs.
- **Emphysema:** It is the abnormal distension of alveoli which results in the loss of elasticity. Cigarette smoking and chronic bronchitis are two main causes.

BIOTECHNOLOGY

- It is the use of microorganisms, their parts or processes for the manufacture of useful or commercial substances. It has two core techniques, i.e. genetic engineering and technique to facilitate the growth and multiplication of only desired microbes.

APPLICATIONS OF BIOTECHNOLOGY

- A number of transgenic plants, medicines, acids are produced through genetic engineering.

BT COTTON

- It was developed to reduce the heavy reliance on pesticides.

BT BRINJAL

- It is a transgenic brinjal created by inserting a crystal protein gene (Cry IAc) from the soil bacterium *Bacillus thuringiensis* into the genome of various brinjal varieties.

GOLDEN RICE

- Golden rice was developed as a fortified food to be used in areas, where there is a shortage of dietary vitamin A.
- **Golden Rice 2** produces up to 23 times more beta-carotene than the original variety of golden rice.

FLAVOUR SAVOUR

- By the use of antisense RNA technology, the enzyme polygalacturonase, which causes damage to pectin is deactivated and the tomato is kept afresh for longer duration.

CANOLA

- It refers to either rapeseed or field mustard. Its seeds are used to produce edible oil suitable for consumption by humans and livestock. The oil is also suitable for use as biodiesel.
- **Yoghurt** is a preserved milk product having a distinct taste and a thick texture than milk.
- **Vitamin C** was the **first vitamin to be produced** by a fermentation process using *Acetobacter,* a wild bacterium.
- The fungus, *Ashbya gossypii* is **used for the microbial production of vitamin B_2.**
- **A biochip** is a discrete collection of gene fragments on a stamp-sized chip that can be used to screen for the presence of particular gene variants.
- Biochips can **help in identifying precise forms of cancer**.
- **Gene therapy** is the treatment of disease by replacing, altering or supplementing a gene that is absent or abnormal and whose absence or abnormality is responsible for the disease.

SOME IMPORTANT FACTS

1. At the time of creation of life there was no oxygen.
2. The strongest part in the body is the enamel of teeth.
3. The fastest nervous speed is 532 kmph.
4. The bones are as strong as concrete and as hard as granite.
5. Inside the body approximately 150 lakh cells are destroyed every second.
6. The weight of the kidney is approximately 150 gram.
7. The blood circulation inside the body takes approximately 23 seconds.
8. The antibiotic, namely penicillin is obtained from *penicillium* fungus.
9. Albatross is the largest sea bird, whose spread of feather is 10-20 ft.
10. In the initial stage of formation of placenta, H.C.G. hormones flow at a large quantity and excreted through urine. At this time, in the testing of urine due to presence of this hormone pregnancy test is carried out.
11. The heartbeat of a child is more than that of an adult.
12. A single respiration completes in five seconds, i.e. two seconds of inspiration and three seconds of expiration.
13. Everyday blood in the body of the human carries approximately 350 liters of oxygen to the cells of the body. Out of this 97% oxygen is carried by haemoglobin and remaining 3% is circulated by blood plasma.

Deficiency Diseases

Deficiency	Diseases	Comments
Vitamin A (retinol)	Xeropthalmia Dermatosis	Lachrimal glands stop producing tears leading to blindness.

(Vitamin B_1) Thiamine	Beri-Beri	Extreme weakness, swelling, pain in legs, loss of appetite, enlarged heart, headache and shortness of breath.
(Vitamin B_2) Riboflavin	Ariboflavinosis	Blurred vision, burning of the eye and tongue, cracking of skin at angle of mouth.
(Vitamin B_3) Niacin (Nicotinamide)	Pellagra,Glossits	Tip and lateral margins of tongue, mouth and gums become red, swollen and develop ulcers
(Vitamin B_5) Pentothenic Acid	Achromotrichia	
Pyridoxine (Vitamin B_6)	Abnormal Protein Metabolism	
(Vitamin B_7) Pantothenic Acid Biotin	Dermatitis, enteritis and anaemia	
(Vitamin B_9) Folic and Folinic Acid	Megaloblast and Birth defects	
(Vitamin B_{12}) Cyanocobalamin	Pernicious or Megaloblastic Anaemia	Reduction of Haemoglobin due to disturbance in the formation of RBC.
Vitamin C (Ascorbic acid)	Scurvy	Pain in joints, loss of weight, gums become spongy and bleed Teeth loose and fragile.
Vitamin D (Cholecalciferol)	Rickets Osteomalacia	Occurs in Children. Softness and deformities of bones Bones susceptible to fracture.
Vitamin E (Tocopherol)		
Vitamin K (Phylloquinone)		
Potassium	Hypokalaemia	Rise in heart beat rate Kidney damage.
Sodium	Hyponatraemia	Low blood pressure.
Proteins	Kwashiorkor	Potbelly due to retention of water by the cells (Oedema).

Diseases caused by Microorganisms

Virus	Bacteria	Protozoas	Fungi	Worms
Small Pox	Sore throat	Malaria	Ringworm	Taeniasis
Chicken Pox	Diphtheria	Amoebic dysentery	Athlete's Foot	Schistosomiasis
Common Cold	Pneumonia	Trypanosomiasis	Madura Foot	Bilharziasis
Influenza/Flu	Tuberculosis	Oriental Sore	Dhobie Itch	Ancylostomiasis
Measles	Plague	Kala Azar		Hookworm
Mumps	Tetanus	Giardiasis		Ascariasis
Encephalitis	Typhoid	Diarrhoea		Enterobiasis
Poliomyelitis	Cholera	Vaginitis		Pinworm disease
Rabies	Bacillary Dysentery			Filariasis
Dengue	Whooping Cough			Elephantiasis
Herpes	Gonorrhoea			
AIDS	Leprosy Botulism			

Hormones

Gland	Hormone	Effect
Pituitary/ Hypophysis Anterior Lobe	Growth Hormone or Somatotrophic Hormone (STH)	Growth of long bones, muscles.
	Thyroid Stimulating Hormone (TSH)	
	Adrenocorticotrophic Hormone (ACTH)	Influences the production of corticosteroids by adrenal cortex involved in defending body against physiological stress.
	Follicle-stimulating Hormone (FSH)	Growth and maturation of follicles in the ovary, production of female sex hormone estrogen and maturation of spermatozoa in males..
	Luteinizing Hormone (LH)	Stimulates interstitial cells in the testis to produce testosterone. Causes ovulation. Release of estrogen & formulation of corpus luteum in female.
	Prolactin or Luteotrophic Hormone (LH)	Helps to maintain pregnancy. Stimulate mammary glands to secrete milk.
Middle Lobe	Melanophore-stimulating Hormone (MSH)	Associated with melaonophyte which gives skin its colour.
Posterior Lobe	Vasopressin or Anti-diuretic Hormone	Controls water reabsorption in the kidney tubule.
	Oxytocin	Causes uterine contractions and active expulsion of milk during and after birth.
Hypothalamus	Releasing Hormone (RH) for each anterior pituitary hormone: GH-RH, TSH-RH, ACTH-RH, FSH-RH and likewise	Production of all the anterior pituitary hormones is controlled by messages from the hypothalamus via hypophyseal portal vessels.
Thyroid	Thyroxine/Calcitonin	BMR influences heat production, Calcium level in blood.
Parathyroids	Parathormone	Raises blood calcium level.
Adrenals	Aldosterone	Regulates sodium and potassium levels in the blood to control blood pressure.
	Hydrocortisone	Plays key role in stress response; increases blood glucose levels and mobilises fat stores; reduces inflammation.
	Epinephrine or Adrenalin	Increases blood pressure, heart and metabolic rate, and blood sugar levels; dilates blood vessels. Also released during exercise
	Norepinephrine/Noradrenalin	Increases blood pressure and heart rate; constricts blood vessels.

Thymus	Thymosin	Development of white blood cells.
Pancreas	Insulin Glucagon	Controls blood sugar level. Increase the blood sugar level
Ovaries	Estrogen	Secondary sexual characteristics.
	Progesterone	Prepares Endometrium (inner lining of Uterus) and maintains it during pregnancy.

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